

**Report
Of
AI-Powered International B2B Sales Automation Platform**

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE (Artificial Intelligence & Machine Learning)



By

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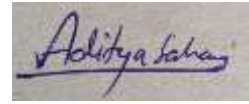
Batch: 2026–27 | Semester 7 | Section 4CSE1

2026–27

CANDIDATE'S DECLARATION

I, Aditya Sahani, do hereby declare that the internship report titled “**AI-Powered International B2B Sales Automation Platform**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science & Engineering (Artificial Intelligence & Machine Learning). All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____



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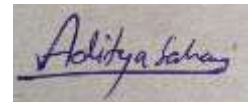
ACKNOWLEDGEMENT

I use this opportunity to express my gratitude to everyone who supported me throughout the Internship Program at **Bonhoeffer Machines Pvt. Ltd.** I am thankful for their guidance valuable constructive criticism and friendly advice throughout the Internship Program. I am sincerely grateful to **Mr. Sahil Hussain** and the **Sales - AI Department** for their mentorship and for the opportunity to work on a production-grade AI project. I would also like to thank Ms. Charu and the HR Department at Bonhoeffer Machines Pvt. Ltd. For their coordination and support throughout the Internship Program.

I would also like to thank the faculty of the **School of Computer Science and Engineering at IILM University in Greater Noida** for their guidance and for giving me the chance to do this industry internship, as part of the curriculum.

I express my gratitude to my parents for their blessings and constant support.

Signature: _____



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INTERNSHIP COMPLETION CERTIFICATE



Certificate of Internship

Date: 25th August 2026

This is to certify that **Mr. Aditya Sahani** has successfully completed **3 months** of Internship with us as **AI & ML Intern** from **21st May 2026** to **21st August 2026**.

During the period of this internship programmed with us he has been exposed to different process, and we found him diligent, hardworking and inquisitive. His contribution towards the task assigned in the organization was "Good".

We wish him every success in his life and career.

From
Bonhoeffler Machines Pvt Ltd.

Bonhoeffler Machines Private Limited

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CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

This report gives an overview of the Summer Internship completed by Aditya Sahani (Enrollment No. CS-2341727) a final-year B.Tech CSE (AI & ML) student from IILM University, Greater Noida as part of the curriculum for the session 2026–27. The Summer Internship took place at Bonhoeffter Machines Pvt. Ltd., Gurugram for a duration of three months starting on 21st May 2026 (Joined 26th May). Ending on August 2026. Aditya Sahani worked in the role of an AI & ML Intern, in the Sales. AI department.

The internship focused on creating, developing and launching an AI-powered international B2B sales automation platform for the company. The tasks included gathering company intelligence setting up automated outreach across channels and building a multilingual conversational AI agent. The experience gave real-life exposure, to production-grade software engineering, managing AI agents and dealing with real-world deployment issues. This report covers the problem the platform solves, the design and tools used the approach taken the difficulties faced and how they were fixed and the results and knowledge gained from the internship.

4.2 Organization Profile

Bonhoeffter Machines Pvt. Ltd. Is a company that makes and sells power equipment for farming, gardens, forests and industry. The company started as Supertec Machines in 1978. Later became Heine Corporation. In 2018 Bonhoeffter Machines was created as the brand for farming and forest equipment.

Now Bonhoeffter Machines is active in than 23 countries. It has than 5,743 distributors and more than 5.7 million customers around the world. The company has two parts. One is the Garden & Forestry Division, which has the brands Bonhoeffter Machines, Stronwell and Mechnova Machines. The other is the Hardware Division, which has the brand Stevron Tools. The products are made in Asia. Follow strict quality rules and are made in places that have ISO certification.

To grow internationally the company wanted to update and use machines to help sell to businesses. This goal was the start of an internship project. The project was, about making a system that uses intelligence. The system helped find customers send

messages in many languages talk to leads manage campaigns and help with the sales process.

4.3 Problem Statement

Before this project Bonhoeffer Machines international sales expansion had operational problems:

- **Manual lead research:** Finding companies that might be interested and the right people to contact in countries like Argentina, Brazil, Bangladesh and Mexico took a lot of time and effort. This made it hard to grow the process.
- **Manual outreach:** Reaching out to new companies through WhatsApp and email was done by hand with each message written and sent individually.
- **Limited follow-up management:** There was no system to track follow-ups automatically. This meant that some leads who did not respond were hard to keep track of and might be missed.
- **Multilingual communication:** Prospects from countries spoke different languages such as Spanish, Portuguese, English and Bengali. This made it hard for the sales team to communicate effectively.
- **Lack of visibility:** There was no one place where the sales team could see all the conversations happening how campaigns were going, how engaged leads were and what information was available about each prospect.
- **Manual meeting coordination:** Scheduling meetings with prospects, from countries and time zones needed a lot of manual work.

4.4 Project Objectives

The primary objectives of the project were to develop an AI-driven system that could streamline and automate Bonhoeffer Machines' international B2B sales and outreach processes:

- **Automate prospect discovery and enrichment:** Automate the identification and enrichment of company and decision-maker contact information for international prospects.
- **Enable multi-channel outreach:** Develop automated WhatsApp and email campaigns capable of reaching and engaging prospects at scale.
- **Develop a multilingual AI conversational agent:** Build an AI agent capable of communicating with international prospects, qualifying leads, sharing

relevant product information, and supporting meeting scheduling with minimal human intervention.

- **Automate lead follow-ups:** Implement a structured, multi-stage follow-up mechanism to ensure that non-responsive leads remain engaged and are not overlooked.
- **Provide centralised CRM visibility:** Develop a unified CRM dashboard to provide visibility into lead information, conversations, campaign activity, and campaign performance.
- **Ensure production readiness and reliability:** Design and deploy a resource-efficient and reliable system capable of operating effectively on limited server infrastructure.

4.5 Scope of the Project

The project was implemented as a two-server platform working in conjunction with one another. The first server, the Company Intelligence Platform, is responsible for researching and enriching company and contact data. The second server, the Global AI Agent Platform, uses this enriched data to run outreach campaigns, host the conversational AI agent (“Valentina”), manage automated follow-ups, schedule meetings, and present all activity through a CRM dashboard. The scope of the internship covered backend development, AI agent design, database architecture, third-party API integrations, frontend dashboard development, and production deployment and maintenance of both platforms.

4.6 Technologies and Tools Used

The following technologies formed the core stack of the Global AI Agent Platform:

Layer	Technology
Backend	Python 3.12, Flask, Gunicorn
AI Framework	CrewAI (multi-agent orchestration)
Large Language Model	OpenAI GPT-4o / GPT-4o-mini
Vector Memory	OpenAI Embeddings + FAISS
WhatsApp Integration	Twilio WhatsApp Business API
Email Integration	Gmail SMTP + Gmail OAuth
Calendar / Meetings	Microsoft Graph API (Teams)

Database	MySQL 8.0, SQLAlchemy
Frontend	React, TypeScript, Vite, Tailwind CSS
Infrastructure	Ubuntu 24, Nginx, systemd
Lead Enrichment APIs	Apollo.io, Hunter.io, Serper.dev

The Company Intelligence Platform additionally used FastAPI with asyncio for asynchronous processing, BeautifulSoup and httpx for web scraping, and a dedicated MySQL database for storing enriched company and contact records with unique Apollo identifiers to prevent duplication.

4.7 System Architecture

The platform uses a two-server design. The Company Intelligence Platform, located at `company.bonhoeffer.online` does research and enrichment. The Global AI Agent Platform, located at `chat.bonhoeffer.online` manages outreach campaigns. Runs the conversational agent. Data that has been enriched flows from the intelligence platform, into the agent platform. In the agent platform this enriched data powers WhatsApp and email campaigns.

4.7.1 Company Intelligence Platform Flow

The Company Intelligence Platform helps to research, enrich and store data about international B2B prospects. The Company Intelligence Platform follows a workflow.

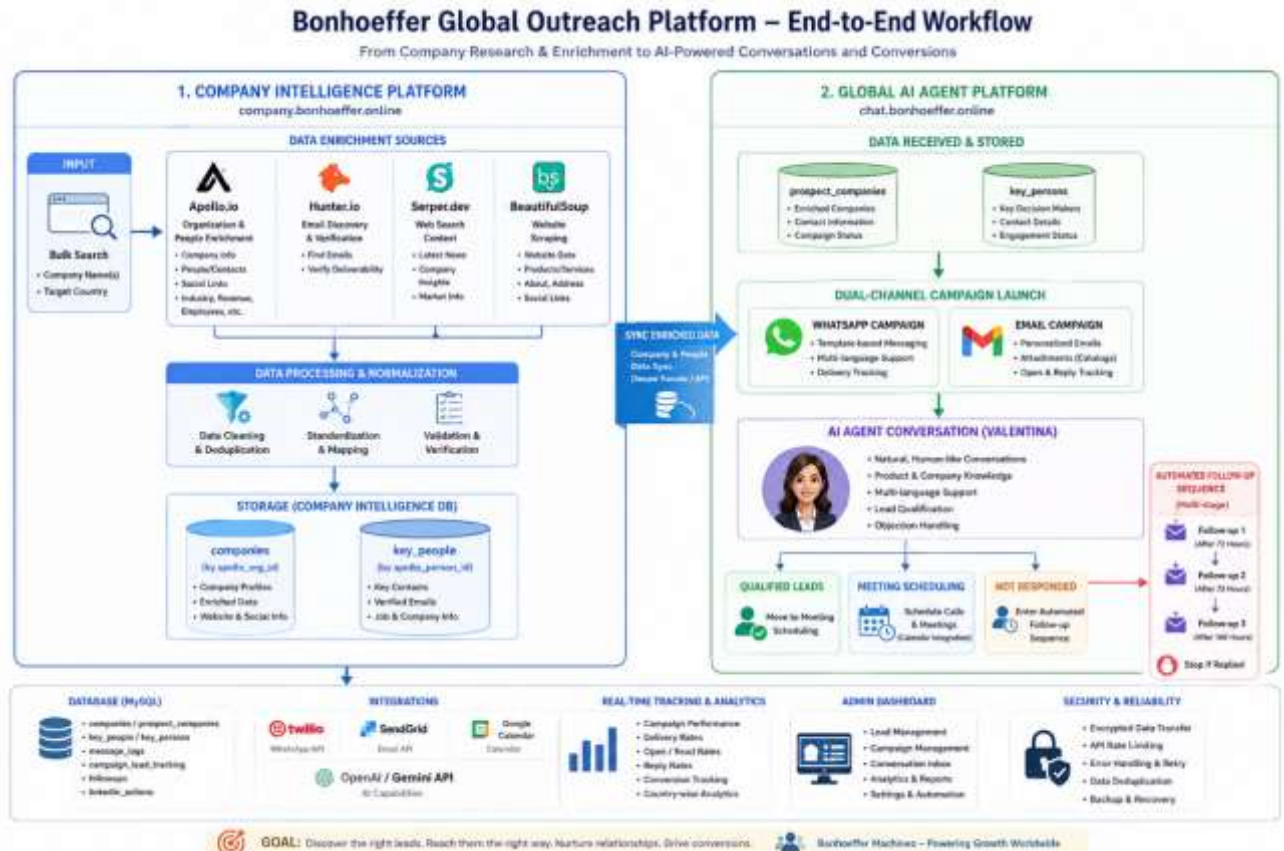
- **Input:** The process starts when you submit a company name and a target country using the bulk-search interface.
- **Company and people enrichment:** The system calls the Apollo.io API to get organisation details and to find the decision-makers inside the company you are looking at.
- **Email enrichment:** Hunter.io helps to find and confirm the email addresses of the key decision-makers when it is needed.
- **Web search and website analysis:** Serper.dev supplies web search data and helps to locate useful company web pages. BeautifulSoup then pulls the available text, from the company site for further study.
- **Data. Deduplication:** The updated company and contact details are saved in the `companies` and `key_people` database tables. The Apollo organisation ID and

Apollo person ID are used as keys so that duplicate entries are avoided and the data stays consistent.

4.7.2 Global AI Agent Platform Flow

The Global AI Agent Platform handles automated -channel contact with better international prospects using the following process:

- Prospect data synchronisation: Better prospect and decision-maker data is put into the prospect_companies and key_persons database tables.
- Multi-channel campaign execution: Selected prospects are put into dual-channel WhatsApp and email outreach campaigns.
- AI-driven conversation handling: Replies from prospects are sent to the Valentina AI agent, which takes care of the rest of the conversation.
- Lead qualification and meeting scheduling: The AI agent talks to prospects finds out what they need checks if they are a lead and if needed helps set up meetings.
- Automated follow-ups: Prospects who do not reply are put into a planned -step follow-up process to keep them involved.



4.8 Methodology

The project was built step, by step with each core feature designed, written, tested in outreach situations and then changed based on how it worked in production. The way each major feature was handled is summarised below.

4.8.1 Valentina – The AI Sales Agent

Valentina is an AI commercial manager built by CrewAI's multi-agent orchestration framework. When a WhatsApp message comes in Valentina routes it through a line of four helper agents. First the Onboarding Agent greets the user in the language it detects and notes the user's role. Second the Intent Analysis Agent looks at the message decides what the user wants—such as a product question, a meeting request or a general inquiry—and pulls out any details. Third the Information Retrieval Agent searches product specs, manuals, FAQs and past chats to find the data. Fourth the Action Executor Agent sends emails books Teams meetings saves information and writes the final reply. Valentina's language detection starts by reading the text of the message. If the text is unclear Valentina falls back to an OpenAI-based detector. Once Valentina decides on a language Valentina remembers that choice, for all conversations.

4.8.2 Dual-Channel Campaign System

Campaigns are started from the dashboard that is built with React. A person picks leads and a WhatsApp message template. They also add an email subject and email message. When the campaign starts, a process that works in the background sends messages to each lead. The messages go to WhatsApp using the Twilio Content Template API. The messages also go to email using Gmail SMTP. The system keeps track of how each message's delivered. It updates the status for each way of sending in the database. The system makes sure there are no duplicates. It uses organisation and person identifiers from Apollo. This means that leads that have already been contacted are not sent messages again when there is a sync.

4.8.3 Automated Follow-Up Sequence

Two follow-up mechanisms were designed. The first mechanism uses a template-based sequence. This sequence starts when a lead does not respond to the campaign. It sends three follow-up messages. These messages are sent at 72-hour and 168-hour intervals. Each message is translated into English, Spanish and Portuguese. The sequence stops

when the lead responds, when a meeting is booked or when the maximum number of steps is reached. The second mechanism uses AI-generated follow-ups. This one activates when a conversation becomes silent after the lead replies. In this case CrewAI reviews the conversation history. It then generates a follow-up message that fits the context. The message is written in the lead's language. A lightweight cron job runs every 15 minutes. It checks for leads that need follow-up. It uses multilingual exit-phrase detection. This helps it stop the sequence if the lead says they are no longer interested.

4.8.4 CRM Dashboard

I built a dashboard using React, TypeScript, Vite and Tailwind CSS to give people operational visibility. The dashboard shows a leads view. The leads view has per-lead delivery-status badges and campaign launch controls. The dashboard also displays a real-time chat view that clearly shows outgoing messages. On the dashboard I added an email-conversation log that integrates with Gmail OAuth. Finally the dashboard offers campaign-management tools so you can select and monitor templates by language.

4.8.5 Key Technical Challenges Resolved

Several production-level engineering challenges were identified and resolved during development:

- Reply deduplication – I noticed that many quick messages from the same user were getting combined into one context with a version-based locking buffer so the agent did not send repeated replies.
- LLM hallucination prevention – I added a guard that checks if the agent says it sent an email. The email was not actually sent. It compares data in the database. Triggers the send if it is missing.
- Tool context injection – I created a wrapper class that safely hands over conversation data like the WhatsApp number and memory status to CrewAI tools because without it the tools got no input.
- Self-healing database schema – I built a check that when the system starts looks at the current database and compares it with the columns that should exist. It then safely adds any parts even if many workers run at the same time.
- Media delivery fix – I solved WhatsApp media failures that happened because Google Drive showed a warning. I switched to a direct-download link.

- Time-zone consistency – I fixed a problem where the database servers local time did not match Python’s UTC-based schedule. This mismatch made follow-up actions fire early. I aligned all scheduling with the database’s time.
- Resource optimisation – I rewrote the follow-up cron job as a independent script that does not load the whole app. This cut memory use from about 500 MB, to 15 MB each run,. The server no longer runs out of resources.

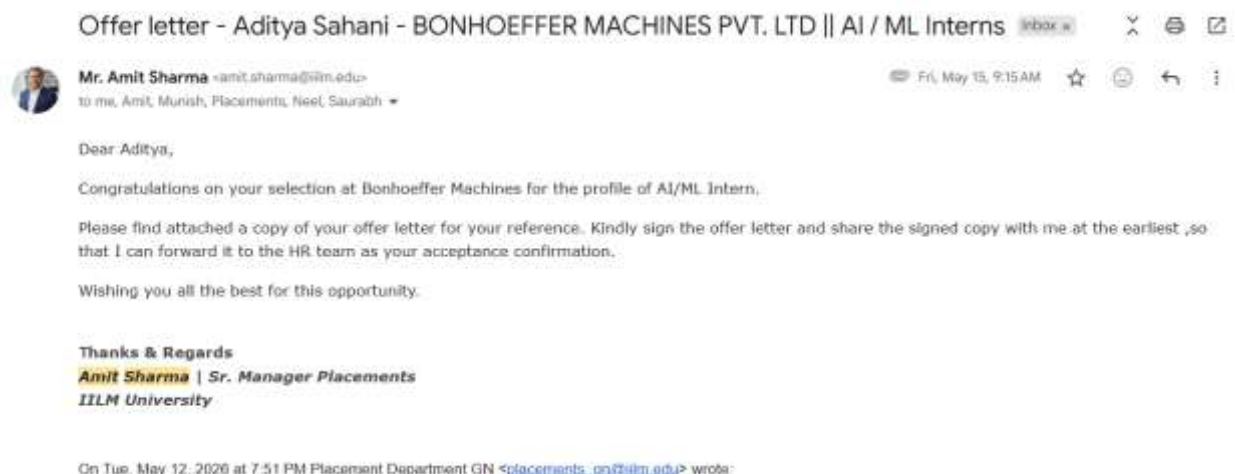
4.9 Expected Outcomes

The finished platform provides the following results for the company:

- Automated lead research that adds information about a company and its main decision makers in minutes of taking hours of work by people.
- Outreach in languages at the same time in English, Spanish and Portuguese across more than 23 countries.
- An AI agent that works around the clock to manage lead conversations without needing people to be involved all the time.
- An automated step-by-step follow-up system that makes sure no lead is missed after the contact.
- A single CRM dashboard that shows everything about each conversation, each campaign, each meeting and each email.
- Automated scheduling of Microsoft Teams meetings, with customers that takes time zones into account.

4.10 Certificates of Completion and Communication Proof

Here is the internship offer letter from Bonhoeffler Machines Pvt. Ltd. It confirms the internship role, the length of the internship and the terms of the internship. The Internship Completion Certificate was already added to this report in Chapter 3. Below is an email screenshot showing the offer letter, from the Placement department.



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